

Sideline Screen / Duck Blind

Construction and Field Operations Manual

The nominal 4 ft x 8 ft sideline screen provides the color guard with a concealed area for storing equipment and making costume or equipment changes. The frame is constructed from 3/4-in. EMT conduit, off-the-shelf brackets, and 3D-printed parts. Decorative vinyl covers the finished face. The completed screen folds flat for transport and storage and unfolds for field deployment.



Screen shown deployed and folded for storage.

Release v0 (unreleased) | Development build D25 | September 1, 2026

Version	Date	Status	One-line description
D24	2026-09-01	Working draft	Imported legacy v24 draft as controlled baseline D24
D25	2026-09-01	Working draft	Standardized status markings, footers, and added visible version history table

Version	Date	Status	One-line description
D26	2026-09-01	Working draft	Standardized document architecture, 2026 CBA rules, transport cart specifications, and canonical cost breakdown tables

Intro / Overview

This master manual is organized as two independently printable appendices under a shared overview. The appendix letter in each header and page number identifies the complete print unit.

Appendix A: Operations Manual

Field-facing copy for transport, deployment, ballasting, post-use inspection, and storage. A field operator should need only the pages labeled A- once the procedures are validated and released.

Appendix B: Construction Manual

Shop-facing copy for materials, cutting, frame assembly, vinyl installation, inspection, and source lookup. The component-source index is located at the end of Appendix B and applies only to that appendix.

Printing and separation

Appendix A print unit: Print or extract every page labeled A-. Its cover identifies the parent manual and revision.

Appendix B print unit: Print or extract every page labeled B-. Its cover and index remain with the construction packet.

Master copy: Keep this Intro / Overview with the complete document so the relationship between both appendices remains clear.

Working-draft status: Appendix A is structurally ready to separate, but its operating procedures are still development notes and must not yet be issued for field use. Appendix B's frame-construction procedure is illustrated; its vinyl-installation procedure remains incomplete.

APPENDIX A

Operations Manual

Field transport, deployment, ballasting, inspection, and storage

This appendix is designed as a complete, independently printable work packet while remaining clearly identified as part of the parent manual.

Audience: Field operators

Contents: Folding and transport; deployment; ballasting; post-use inspection and storage

Print or carry: All pages labeled A-

Parent revision: Release v0 (unreleased) | Development build D26 | September 1, 2026

WORKING DRAFT - NOT READY FOR FIELD ISSUE: Deployment sequence, approved ballast quantity, and operating limits still require validation. Do not issue this appendix to field crews yet.

When separated from the master document, keep this cover as the first page of Appendix A.

Field Operations

Appendix A is the field-facing portion of the manual. It is designed to be printed and carried independently by operators responsible for transport, deployment, inspection, ballasting, and storage.

Section status: The procedures below are development notes, not a complete field-use procedure. Deployment sequence, approved ballast quantity, and operating limits still require validation.

Folding and transport

To prepare sideline screens for transport, handlers release the hinged arm clips, fold the bottom support arms (E) upward against the rear frame, and collapse the entire rear support frame flat against the main frame. Multiple folded screens are loaded vertically onto a dedicated rolling transport cart between side guide rails. Front and rear retaining gates are closed and latched (Cart Gate Latch) to lock screens in place during transit from the equipment trailer to the stadium entrance.

Deployment and Sideline Cart Stowage

At the performance field gate, the transport cart is rolled to the sideline staging position. Handlers unlatch and open the front and back cart gates, lift individual folded screens from the cart bed, and carry them to their designated sideline coordinates. Once on coordinate, operators swing the rear support frame open and snap the hinged arm clips securely over the bottom support arms (E). During the performance, the empty transport cart remains parked and braked along the sideline outside the performance boundary. Immediately upon the final performance cue, screens are collapsed flat, rapidly reloaded into the transport cart, gates latched, and the cart rolled off the field in continuous movement per Rule 8.05.

Ballasting

The current design allows ballast to be suspended from upper weight clips. Placing weights over the bottom support components may fatigue or break the 3D-printed hinged arm clips and is not an approved method until load testing is complete.

Do not issue for field use: Before this subsection is released, specify the approved weight clips, ballast quantity, attachment locations, wind limits, inspection criteria, and response to changing weather.

Applicable 2026 Marching Band Competition & Prop Rules

The following governing rules are extracted directly from the official 2026 Colorado Bandmasters Association (CBA) Marching Band Rulebook (Appendix B Index [11]). All prop builders, parent crew, and student field operators must adhere to these standards:

ANNUAL RULEBOOK NOTICE: This section reflects the governing rules for the 2026 competitive season. Logistics staff and build leads must review this section annually against updated CBA and Bands of America (BOA) rulebooks to verify continuing compliance.

Rule 8.05: Props, Equipment, Surface Protection, and Double-Bagging

- **Gate Ingress:** All props and equipment must be designed and of a quantity such that they can be brought onto the Performance Field from the band entrance gate.
- **Continuous Movement:** Following the end of the band's Performance, all props and equipment must be in continuous movement until entirely removed from the stadium.
- **Turf Contact Protection:** All wooden props must be protected with a heavy-duty sustainable plastic product (PVC, Melamine, etc.) where said prop comes in contact with the field surface. Steel props with smooth edges are acceptable without the need for additional protection.
- **Secondary Containment for Sandbags:** Props must not leave holes in the Performance Field surface. Sand bags must be in a secondary container (bucket, double bagged, etc.) that will prevent sand from leaking.

Rule 8.07: Equipment Wheel and Turf Protection Standards

- **Pneumatic-Like Tires:** All instruments and equipment wheeled into the stadium from the entrance gate forward must have pneumatic-like tires. Wheels must be able to support the weight of the instrument/prop without creating damage to the field surface.

Rule 8.08: Prop Staging Height Restrictions

- **12-Foot Rigid Height Limit:** Staging (props, backdrops, screens, or similar objects) built and/or used by bands at CBA sanctioned events shall be limited to a maximum total height of twelve (12) feet, including wheels, platforms, and safety railings. Materials such as wood, metal, plastic, or aluminum are not permitted above the twelve-foot limit. (Note: The Sideline Screen is a nominal 4 ft height when deployed, safely conforming to this rule).

Rule 8.09: Prop Assembly, Timing, and USAFA Falcon Stadium Clearance

- **Staging Ingress:** A band may bring props into the stadium after the previous band has entered the field. Props must fit in the end zone curve without blocking entrance/exit gates.

- **Falcon Stadium 9'6" Tunnel Rule:** Any prop being used at Falcon Stadium (United States Air Force Academy) must not exceed 9 ft 6 in. to clear the tunnel entrance. Props up to the 12-ft limit may be erected in the tunnel after clearing the initial entrance.
- **Falcon Stadium Venue Update (2025/2026):** While the 9'6" tunnel restriction remains in the printed CBA rulebook, it has been rendered obsolete by stadium renovations completed prior to the 2025 season. The lower overhead crossbeam at the bottom of the Falcon Stadium tunnel was permanently removed, allowing props to clear without restriction.

Rule 9.07: Parent Field Access & Wristband Limits

- **25 Field Pass Wristbands:** Access to the field is restricted to Directors/Staff with CBA passes and parents assisting with props/front ensemble with designated Field Pass wristbands. Each band receives exactly 25 Field Pass wristbands upon check-in. Additional wristbands are not available.

Post-use inspection and storage

This subsection will cover ballast removal, damage checks, drying and cleaning, folding, transport, storage position, and reporting damaged brackets, conduit, vinyl, or printed parts.

APPENDIX B

Construction Manual

Materials, cutting, frame assembly, vinyl installation, and source lookup

This appendix is designed as a complete, independently printable work packet while remaining clearly identified as part of the parent manual.

Audience: Fabrication and installation crews

Contents: Components; tools; cutting; frame assembly; vinyl installation; Appendix B Index

Print or carry: All pages labeled B-

Parent revision: Release v0 (unreleased) | Development build D26 | September 1, 2026

WORKING DRAFT - VINYL PROCEDURE PENDING: The illustrated frame-construction procedure is included. The vinyl-installation procedure remains reserved for a future development pass.

When separated from the master document, keep this cover as the first page of Appendix B.

Construction

Appendix B is for the fabrication and installation crew. It covers one complete frame and reserves a separate section for vinyl installation. Field operations are maintained independently in Appendix A.

Orientation rule: The finished face is the side that will be covered by decorative vinyl. Keep self-drilling screw heads, T-bracket nuts, bolt tails, and other sharp projections on the rear face unless a step specifically directs otherwise.

Safety and work area

1. Wear eye protection when cutting conduit or driving self-drilling screws. Wear gloves while handling cut conduit.
2. Cut ends can be sharp. Deburr every cut before assembly.
3. Assemble the frame on a large, flat surface, such as a garage floor.
4. Check the frame for square frequently. The brackets allow some movement, so correct misalignment when practical without forcing the assembly.
5. After the outer frame is complete, install each successive rail or bracket toward the center of the frame, immediately inside the previously installed component.

Materials and components

Purchase seven 10-ft lengths of 3/4-in. EMT conduit for one screen. Cut them into components A-E using the optimized cut plan below. The Index ref. column links each component to its purchase source, print material, or digital part files in the Appendix B Index at the end of this appendix. The illustrated inventory that follows provides a visual identification check for components A-J.

ID	Qty.	Component	Specification	Index ref.
A	4	Long main-frame rails	92.5-in. EMT conduit	1
B	2	Main-frame end rails	44.5-in. EMT conduit	1
C	1	Rear cross rail	86.5-in. EMT conduit	1
D	2	Rear-frame arms	37-in. EMT conduit	1
E	2	Bottom support arms	27.5-in. EMT conduit	1
F	6	Corner brackets	Purchased clamp brackets	2

ID	Qty.	Component	Specification	Index ref.
G	8	T-brackets	Purchased clamp brackets; four serve as hinges	3
H	6	Corner plugs	3D printed in black ASA	5 , 6
I	2	Hinged arm clips	3D printed in black ASA	5 , 7
J	36	Self-drilling sheet-metal screws	#8 x 1/2-in. external-hex head	4

Illustrated component inventory

Match the letter on each drawing to the ID in the component table before cutting or assembly.

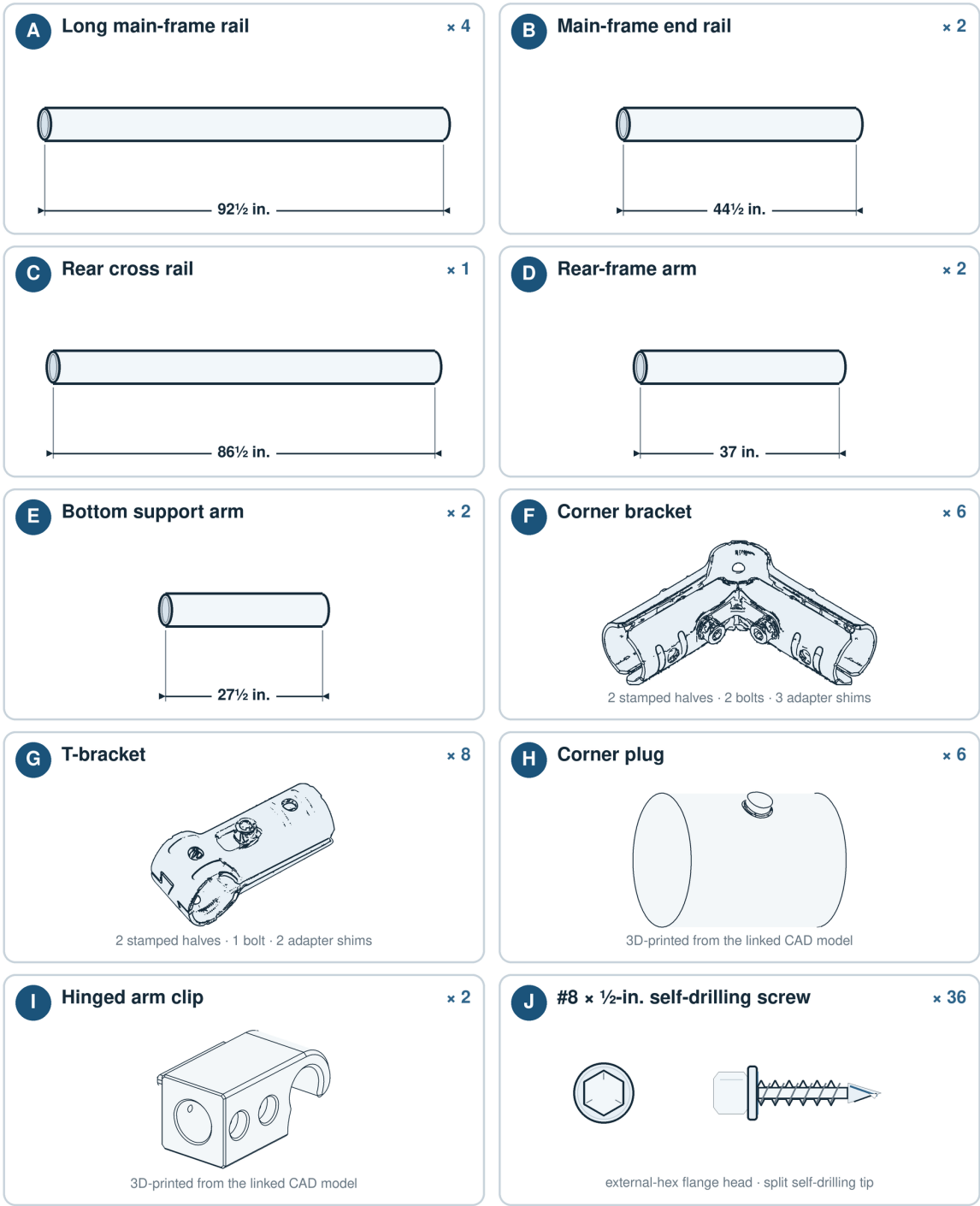


Figure 1. Component inventory for one complete frame.

Identification note: Drawings are not measuring templates and items are not shown at a common scale. The written dimensions and quantities control.

Tools

6. EMT conduit cutter
7. Tape measure and permanent marker
8. Framing square or a tape measure long enough to compare frame diagonals
9. Deburring tool or metal file
10. 5 mm Allen key
11. Drill/driver with a 3/8-in. nut-driver bit
12. Eye protection and work gloves
13. 3D printer capable of printing ASA, unless the printed parts are prepared in advance

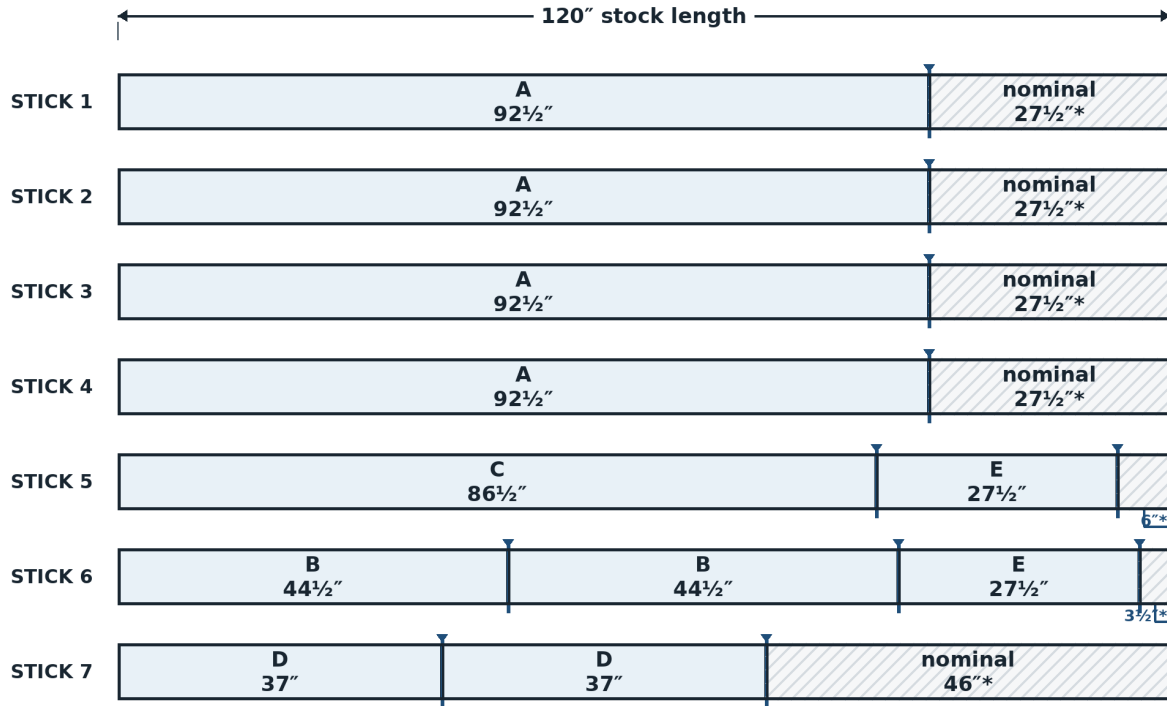
3D-printed parts

Black ASA is the recommended material. Polymaker ASA has provided adequate strength and UV stability in band-prop service for three seasons. PETG may be substituted, but strength varies substantially by brand and formulation; print and load-test a sample before producing all structural parts. Previous Sunlu PETG feather bases fractured in service.

Material restriction: Do not use PLA or ABS for these parts because neither is suitable for prolonged outdoor UV exposure.

Optimized conduit cut plan

The following plan produces every required tube from seven 10-ft stock lengths and does not depend on the nominal 27.5-in. remainder from a 92.5-in. cut. This leaves allowance for normal cutting loss. Keep every finished piece within 1/8 in. of its specified length.



* NOMINAL OFFCUT BEFORE CUTTING LOSS — SEE CUTTING NOTE BELOW

Figure 2. Proportional seven-stick conduit cut diagram.

Cutting note: Gray areas are nominal remainders before cutting loss. Measure every retained piece from a clean stock end. Do not use the nominal 27.5-in. remainder after an A cut as an E arm.

Detailed cut list

Stock length	Retained cuts	Approximate remainder
1	1 x 92.5 in. (A)	About 27.5 in.; spare/offcut
2	1 x 92.5 in. (A)	About 27.5 in.; spare/offcut
3	1 x 92.5 in. (A)	About 27.5 in.; spare/offcut
4	1 x 92.5 in. (A)	About 27.5 in.; spare/offcut
5	1 x 86.5 in. (C) + 1 x 27.5 in. (E)	6 in., less cutting loss
6	2 x 44.5 in. (B) + 1 x 27.5 in. (E)	3.5 in., less cutting loss
7	2 x 37 in. (D)	46 in., less cutting loss

Cutting procedure

1. Label the seven 10-ft stock lengths 1-7.
2. Mark every retained cut before cutting. Verify the complete cut plan before starting.
3. Cut each tube to length. Measure the finished piece rather than assuming an offcut is exact.
4. Deburr the inside and outside edges of every cut.
5. Label the finished tubes A-E so they can be identified during assembly.

When building multiple screens, pool and label usable offcuts before purchasing additional stock.

Estimated fabrication cost breakdown

The following tables itemize estimated material costs per sideline screen and for the planned rolling transport cart based on standard off-the-shelf retail hardware pricing (excluding custom printed vinyl):

Per Sideline Screen / Duck Blind (Excl. Vinyl)

Component / Category	Items Included	Est. Cost
Conduit Framing	7x 3/4" x 10-ft EMT conduit sticks (Home Depot SKU 0550110000)	\$40.50
Corner Brackets (F)	6x 3/4" 3-way corner brackets (Amazon B0D5BBKCKG)	\$27.00
T-Brackets (G)	8x 3/4" EMT T-brackets (Amazon B0CKQW11SJ)	\$23.00
Fasteners (J)	36x #8 x 1/2" hex flange self-drilling screws	\$4.00
3D-Printed Parts	6x Plugs H, 2x Clips I, 2x Weight Clips in Black ASA	\$10.00

Component / Category	Items Included	Est. Cost
TOTAL PER SCREEN	Complete folding screen frame	~\$105.00

Sideline Screen Transport Cart (Component 2 Estimate)

Subsystem / Category	Items Included	Est. Cost
Rolling Base Chassis	Backdrop-spec 2x4 framing, 1/2" plywood deck, 4x casters w/ brakes, paint	\$155.00
Side Guide Rails	Vertical framing / conduit dividers to hold screens upright	\$35.00
Front & Rear Retention Gates	Hinged gate framing, strap/barrel hinges, and 3D-printed latches	\$30.00
TOTAL PER TRANSPORT CART	Complete rolling transport cart	~\$220.00

Frame assembly

Complete the four stages in order. At every hinge location, fasten the bracket to the short moving tube only; never drive a screw into the long rail around which the bracket must rotate.

Folded-state connection map

Use Figure 3 to identify the finished folded arrangement, then use Figure 3A to see how the individual members and connectors align before beginning Stage 1. Both drawings use the same component labels and connector color key. Tube width and separation distances are enlarged for clarity.

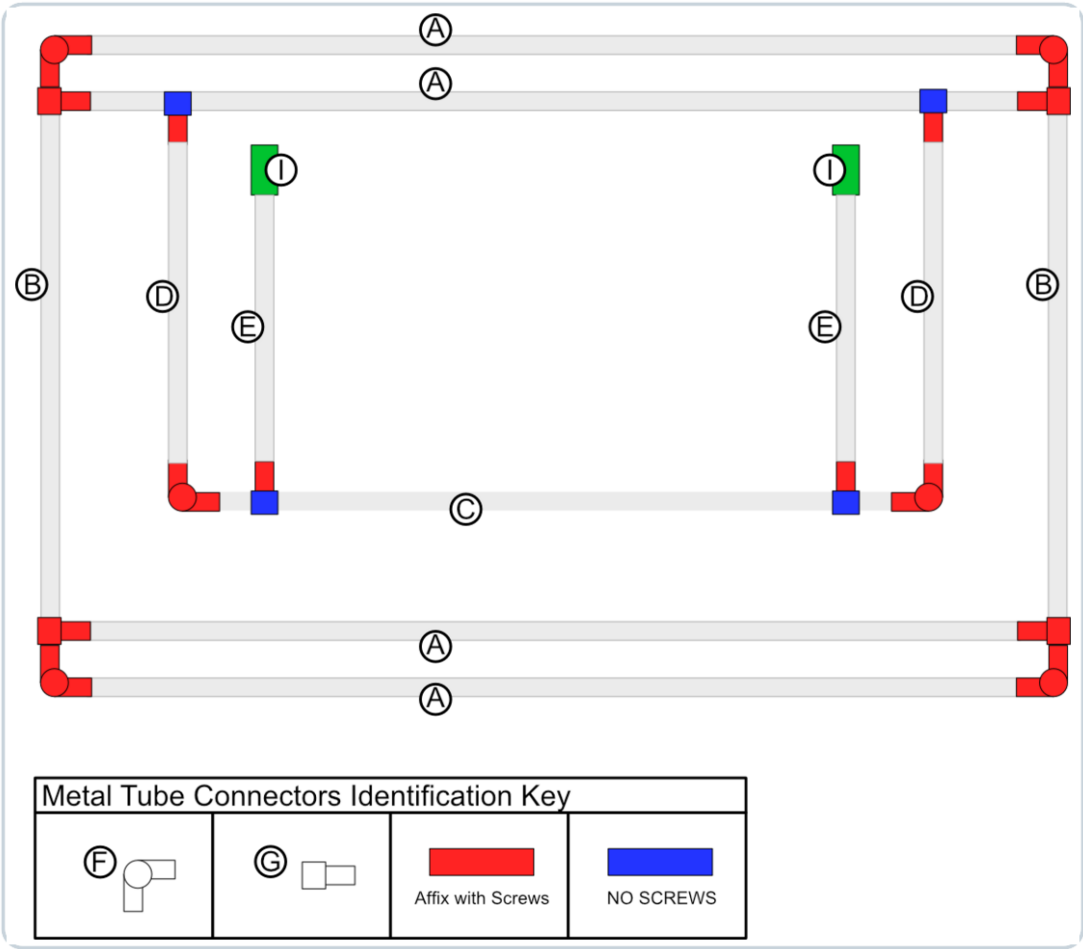


Figure 3. Folded-state component and connector-fastening map for the finished frame.

Exploded folded-state component view

Use this exploded view to identify every construction component in the folded assembly. The separated member ends show how the A-E tubes align with the F and G connectors. Red markers identify connections that receive Item J screws, blue markers identify interfaces that must remain unfastened to pivot, and green blocks identify the two Item I hinged arm clips. Individual Item J fasteners are intentionally omitted.

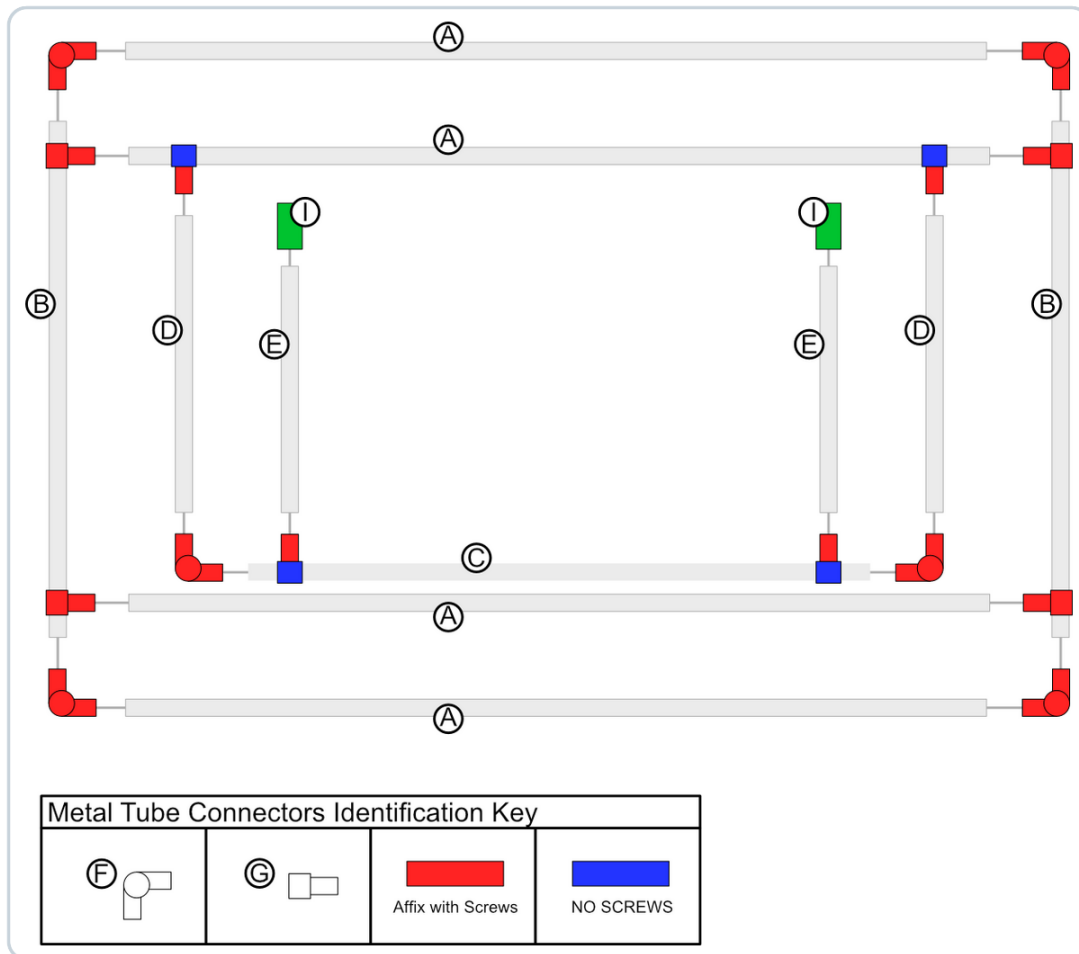


Figure 3A. Exploded folded-state component and connector-fastening map.

Connection rules

Affix with screws — At every red connector marker, secure the connection from the inside or rear face with Item J self-drilling screws, using the quantity and locations specified in the applicable build stage.

No screws — At every blue connector marker, do not drive an Item J screw into the tube that must remain free to pivot. Install the specified screws only on the red-marked fixed side of the connector.

Protect the vinyl — Orient every Item J screw head, T-bracket nut, and bolt tail toward the inside or rear of the assembly, away from the finished face and vinyl.

Stage 1: Assemble the main frame

Parts for this stage

6. 2 x A rails
 7. 2 x B rails
 8. 4 x F corner brackets
 9. 4 x H corner plugs
 10. 8 x J screws
1. Lay two A rails and two B rails on the flat work surface to form a rectangle.
 2. Join the tubes with four F corner brackets. Insert one H corner plug into each unused third socket, rotating it so the side peg faces the rear and aligns with the hole on the bracket's outside bend.
 3. Measure both diagonals. Adjust the frame until the diagonal measurements are equal, then tighten all bracket-retaining bolts.
 4. Install one J screw at each of the eight tube-to-bracket joints. Drive the screws from the inside edge of the rectangle so no screw head faces the finished face.

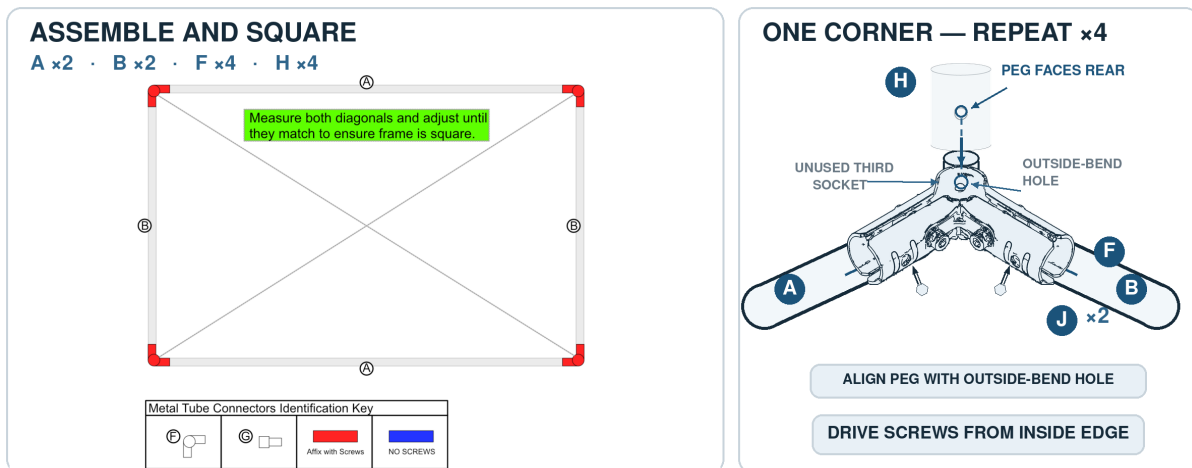


Figure 4. Stage 1 main-frame assembly and corner detail.

Stage 2: Install the inner cross rails

Parts for this stage

5. 2 x A rails
6. 4 x G T-brackets
7. 8 x J screws

Fastener orientation: Place the T-bracket nuts and bolt tails on the rear face. The front/finished face must remain free of sharp projections that could damage the vinyl.

1. Place one A rail immediately inside the top A rail and the other A rail immediately inside the bottom A rail.
2. Attach each end of both A rails with a G T-bracket. Insert each bracket bolt from the finished face toward the rear so the nut is on the rear face, then tighten fully.
3. Flip the main frame so the rear face is up.
4. Drive one J screw through each of the eight tube-to-T-bracket interfaces from the rear face.

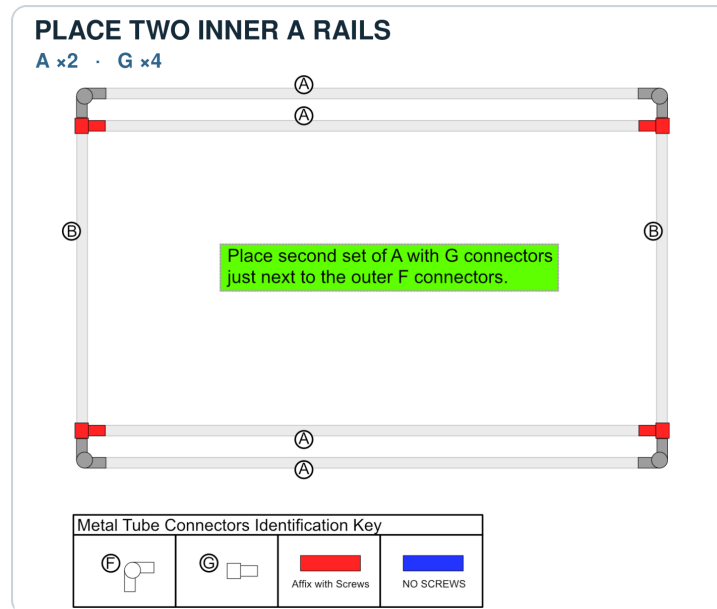


Figure 5. Stage 2 inner-rail placement and T-bracket fastening.

Stage 3: Assemble and attach the rear support frame

Parts for this stage

5. 2 x D arms
6. 1 x C rear cross rail
7. 2 x F corner brackets
8. 2 x G T-brackets
9. 2 x H corner plugs
10. 8 x J screws

Critical hinge rule: Each G bracket in this stage must pivot around the upper A rail. Install two screws in each D arm, but no screws in the A rail.

1. Flip the main frame so the finished face is up. No additional self-drilling screws will be driven through the finished face.
2. Choose either long edge as the top and mark it. The upper inner cross rail is the second A rail from the top.
3. Attach one D arm near each end of the upper inner cross rail with a G T-bracket. Position each new bracket immediately inside the previously installed bracket at that end.
4. At each G T-bracket, drive two J screws into the D arm only (four screws total). Do not drive screws into the A rail; the bracket must rotate freely around that rail.
5. Join the free ends of the two D arms with the C rear cross rail, two F corner brackets, and two H corner plugs.
6. Square the rear support frame and tighten both corner brackets.
7. Install two J screws at each rear-frame corner (four screws total), placing them on the inside of the corner. Stage total: eight screws.

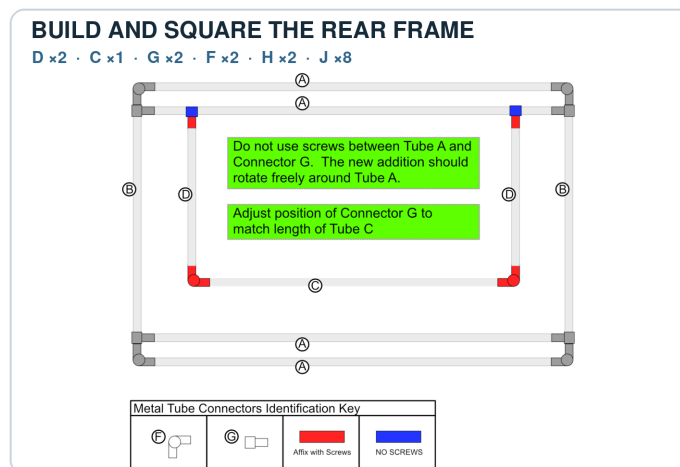


Figure 6. Stage 3 rear-support-frame assembly and hinge fastening.

Stage 4: Install the bottom support arms

Parts for this stage

8. 2 x E support arms
9. 2 x G T-brackets
10. 2 x I hinged arm clips
11. 12 x J screws

Critical hinge rule: Each G bracket in this stage must pivot around the C rear cross rail. Install screws in the E arms only; do not fasten either bracket to the C rail.

1. Attach one E support arm near each end of the C rear cross rail with a G T-bracket. Position each new bracket immediately inside the adjacent corner bracket and tighten the bracket bolts.
2. At each G T-bracket, drive two J screws into the E arm only (four screws total). Do not drive screws into the C rail; the bracket must rotate freely around that rail.
3. Stand the assembly upright with both E support arms lying on the ground.
4. Fit one I hinged arm clip onto the free end of each E arm. Engage both clips with the lower inner cross rail, which is the third long rail from the top and the second from the bottom.
5. Make each clip square to the lower inner cross rail and confirm that it is fully seated on its E arm.
6. Drive four J screws through each hinged arm clip into its E arm (eight screws total). Start with the accessible outside pair, disengage the clip, and install the remaining two screws. Stage total: twelve screws.

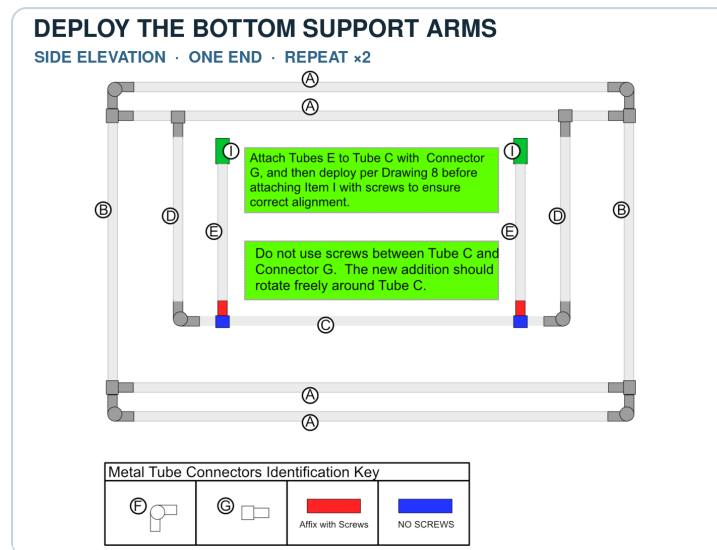


Figure 7. Stage 4 bottom-support-arm installation and clip fastening.

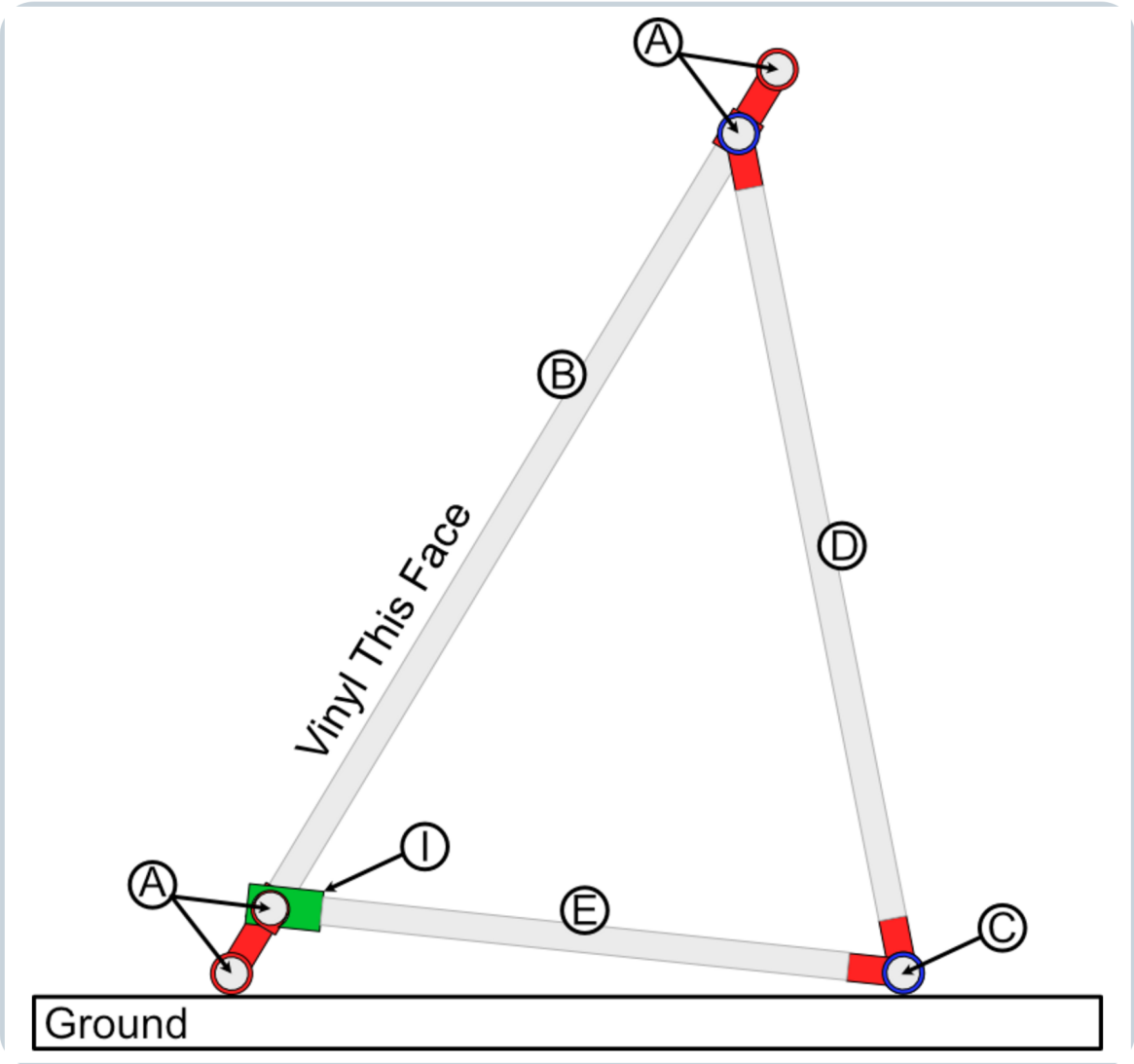


Figure 8. Deployed cutaway side elevation showing the B foot and C/E connection.

Final construction inspection

7. All 36 self-drilling screws are installed in the specified locations.
8. No screw head, bracket nut, bolt tail, burr, or other sharp projection faces the future vinyl surface.
9. Both rear-frame hinges rotate around the upper inner rail without binding.
10. Both bottom-support hinges rotate around the rear cross rail without binding.
11. Both hinged arm clips engage and disengage without excessive force.
12. All retaining bolts are tight, and every structural corner is as square as practical.
13. The frame can be unfolded, locked open with both clips, released, and folded flat.

Transport Cart Fabrication (Planned Component)

To support efficient field transport and rapid deployment during competitive shows, a dedicated rolling transport cart will be constructed to hold multiple folded sideline screens / duck blinds:

- **Backdrop Cart Base Compatibility:** The transport cart utilizes the same heavy-duty 96.0 in. x 44.5 in. 2x4 lumber base framing, 1/2-in. plywood decking, and four heavy-duty swivel plate casters standardized for the Rolling Backdrop System. Building on this shared chassis ensures dimensional consistency and uniform trailer nose wall loading.
- **Side Guide Rails:** Vertical guide rails are mounted along both the left and right longitudinal edges of the cart bed to contain folded sideline screens standing upright in parallel slots during rolling transit.
- **Front & Rear Latching Gates:** Hinged retention gates are installed across the front and rear openings of the cart. The gates are secured with 3D-printed Cart Gate Latches (Cart Gate Latch.FCStd) to lock screens firmly inside the cart bed while allowing quick release and unobstructed unloading at the field sideline.
- **Status:** Provisional design specification. Detailed CAD models, cut lists, and fastener schedules for the transport cart modifications will be finalized in subsequent build releases following prototype testing.

Vinyl Installation

Section status: Reserved for the next development pass. This working draft does not yet provide a complete vinyl-installation procedure.

This section will document vinyl dimensions, cutting, finished-face orientation, attachment hardware and spacing, tensioning and squaring, edge protection, inspection, repair, and replacement.

Appendix B Index

This index applies to Appendix B. Use the numbered references in the Appendix B component table to locate purchase sources, print material, and digital part files. In the digital manual, select a descriptive link. For a printed copy, the complete URL is shown beneath each link.

Purchase sources

[1] [3/4-in. x 10-ft EMT conduit - Home Depot](#)

<https://www.homedepot.com/p/3-4-in-x-10-ft-Electrical-Metallic-Tubing-EMT-Conduit-0550110000/202068040>

[2] [Corner brackets - Amazon](#)

<https://www.amazon.com/dp/B0D5BBKCKG>

[3] [T-brackets - Amazon](#)

<https://www.amazon.com/dp/B0CKQW11SJ>

[4] [#8 x 1/2-in. self-drilling screws - Home Depot](#)

<https://www.homedepot.com/p/Teks-8-x-1-2-in-External-Hex-Flange-Hex-Head-Self-Drilling-Screws-280-Pack-21308/100188836>

[5] [Black ASA filament - Amazon](#)

<https://www.amazon.com/dp/B09TR84J5M>

Digital part files

[6] Corner plug digital files

[FreeCAD source file](#)

https://github.com/ericcrowe/music/blob/7ab68c66b4ecbd72d9e039212acab02a002df1bb/PCHSMB/_Sideline%20Screen/Corner%20Plug.FCStd

[STEP file](#)

https://github.com/ericcrowe/music/blob/7ab68c66b4ecbd72d9e039212acab02a002df1bb/PCHSMB/_Sideline%20Screen/STEP/Corner%20Plug-Part.step

[3MF print file](#)

https://github.com/ericcrowe/music/blob/7ab68c66b4ecbd72d9e039212acab02a002df1bb/PCHSMB/_Sideline%20Screen/3MF/Corner%20Plug-Part.step.3mf

[7] Hinged arm clip digital files

[FreeCAD source file](#)

https://github.com/ericcrowe/music/blob/7ab68c66b4ecbd72d9e039212acab02a002df1bb/PCHSMB/_Sideline%20Screen/Hinged%20Arm%20Clip.FCStd

[STEP file](#)

https://github.com/ericcrowe/music/blob/7ab68c66b4ecbd72d9e039212acab02a002df1bb/PCHSMB/_Sideline%20Screen/STEP/Hinged%20Arm%20Clip-Part001.step

[3MF print file](#)

https://github.com/ericcrowe/music/blob/7ab68c66b4ecbd72d9e039212acab02a002df1bb/PCHSMB/_Sideline%20Screen/3MF/Hinged%20Arm%20Clip-Part001.3mf

[11] [2026 Colorado Bandmasters Association \(CBA\) Marching Band Rulebook \(PDF\)](https://ebf5c7e8-3869-4325-bd98-ce10f57d7545.filesusr.com/ugd/83f67e_cb5fe6efdbdb4ce98f619580f694bc94.pdf)

https://ebf5c7e8-3869-4325-bd98-ce10f57d7545.filesusr.com/ugd/83f67e_cb5fe6efdbdb4ce98f619580f694bc94.pdf